

Questions on P110 needed for algorithms

1. Overall Dimensions
 - Bounding box or radius
 - Length, width, height
2. Payload
 - Payload size & location
 - Max payload configuration geometry
3. Center of Mass
 - Nominal center of mass
 - Center of mass shift with payload
4. Structural clearance constraints
 - Areas that must never get near the obstacles

*Needed for Obstacle inflation radius, minimum corridor width, clearance safety margins, and configuration space modeling

Structures Answers

Bounding Box Dimensions:

Length = Approximately 14.5 ft (12.5 ft structure plus 1.75 ft propeller sticking out the front)

Width/Wingspan = 15 ft

Height = Approximately 4 ft (Subject to change based on landing gear height)

Payload Box Dimensions:

Length = 6 ft

Width = 0.75 ft

Height = 0.75 ft

Center of Mass:

Currently in the Process (Previously around 4-4.1 ft, starting from the vertical propeller tip in the front of the aircraft (frontmost object of our aircraft))

Structural Clearance Constraints:

Main Wing Tips Possibly (Dont want to damage main wing and possibly forward motors/props)

Not sure on the rest